EXPERIMENT 3

clc; clear; close all;

% Parameters fs = 100000; % Sampling Frequency t = 0:1/fs:0.02; % Time Vector fm = 500; % Message Frequency fc = 5000; % Carrier Frequency

% Message Signal m = cos(2*pi*fm*t);

% Carrier Signal c = cos(2*pi*fc*t);

% DSB-SC Modulation dsb = m .* c;

% VSB Filter (Low-pass Approximation) [b,a] = butter(4, fc/(fs/2)); vsb = filter(b,a,dsb);

% Coherent Demodulation demod = vsb .* (2*cos(2*pi*fc*t));

% Recover Message [b1,a1] = butter(4, 1000/(fs/2)); rec = filter(b1,a1,demod);

% Display Results figure;

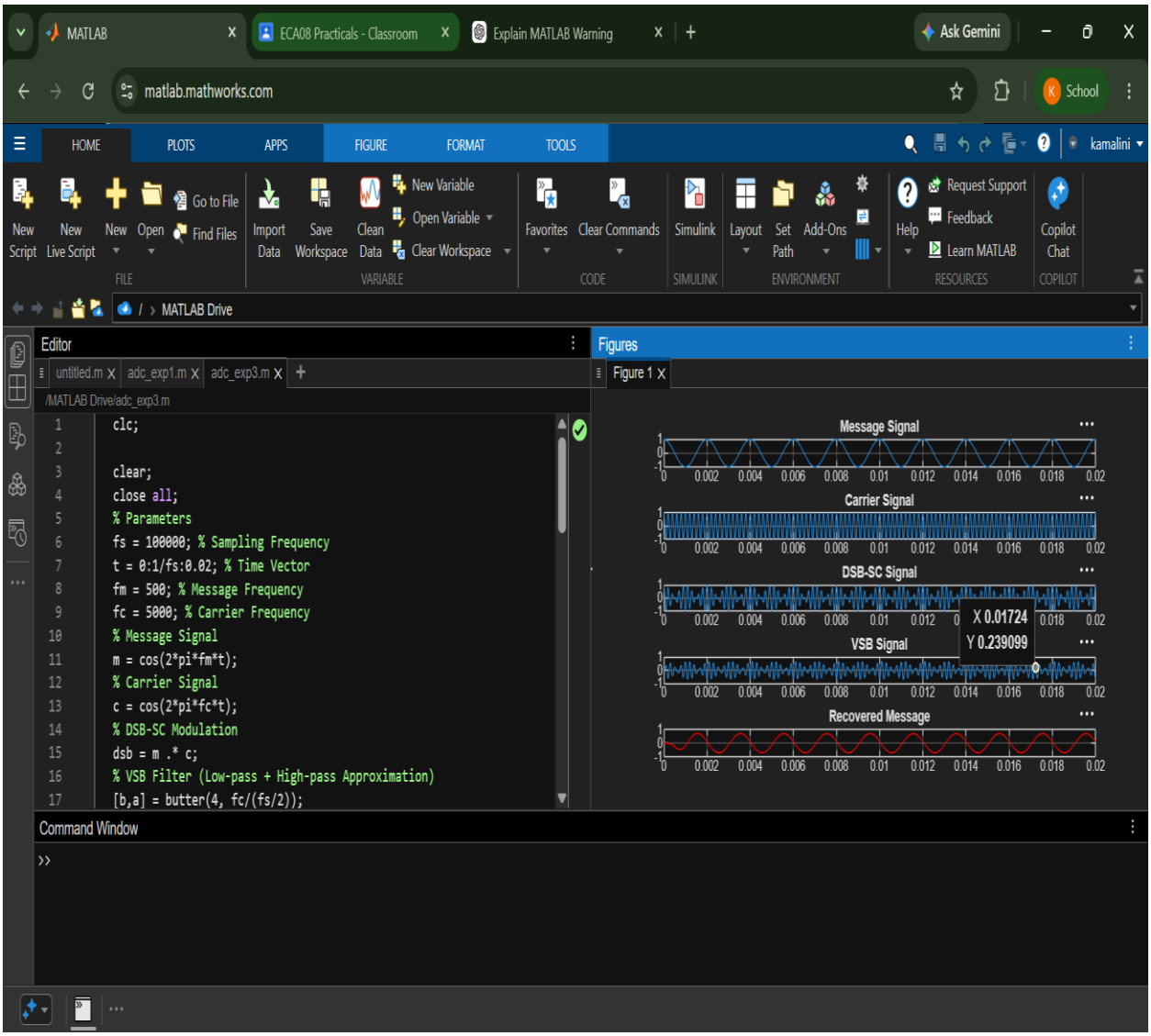
subplot(5,1,1); plot(t,m); title('Message Signal'); grid on;

subplot(5,1,2); plot(t,c); title('Carrier Signal'); grid on;

subplot(5,1,3); plot(t,dsb); title('DSB-SC Signal'); grid on;

subplot(5,1,4); plot(t,vsb); title('VSB Signal'); grid on;

subplot(5,1,5); plot(t,rec,'r'); title('Recovered Message'); grid on;


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FILE
VARIABLE
CODE
SIMULINK
ENVIRONMENT
RESOURCES
COPILOT
MATLAB Drive
Editor
untitled.m
adc_exp1.m
adc_exp3.m
/MATLAB Drive/adc_exp3.m
clc;
clear;
close all;
% Parameters
fs = 100000; % Sampling Frequency
t = 0:1/fs:0.02; % Time Vector
fm = 500; % Message Frequency
fc = 5000; % Carrier Frequency
% Message Signal
m = cos(2*pi*fm*t);
% Carrier Signal
c = cos(2*pi*fc*t);
% DSB-SC Modulation
dsb = m .* c;
% VSB Filter (Low-pass + High-pass Approximation)
[b,a] = butter(4, fc/(fs/2));
Figures
Figure 1
Message Signal
Carrier Signal
DSB-SC Signal
X 0.01724
Y 0.239099
VSB Signal
Recovered Message
0
0.002
0.004
0.006
0.008
0.01
0.012
0.014
0.016
0.018
0.02
Command Window